

# Crypto Daytrading Arena

## Windows Setup Guide

### Prerequisites

Install these three tools before starting. All are free.

|                             |                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Python 3.10+</b>         | <a href="https://python.org/downloads">https://python.org/downloads</a> -- Check "Add Python to PATH" during install                      |
| <b>uv (package manager)</b> | Open PowerShell, run: <code>powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1   iex"</code>                     |
| <b>Docker Desktop</b>       | <a href="https://docker.com/products/docker-desktop">https://docker.com/products/docker-desktop</a> -- Enable WSL 2 backend when prompted |

### STEP 1 Extract & Install

Extract crypto-daytrading-arena.zip to a folder (e.g. C:\arena), open PowerShell, and install dependencies:

```
cd C:\arena
uv sync
```

### STEP 2 Launch the Arena

Run the interactive launcher. It starts Kafka automatically and walks you through configuration:

```
uv run python launcher.py
```

The wizard will ask you to configure:

- Trading mode -- Simulated (paper trading) or Live (real Coinbase orders)
- LLM provider -- Enter your OpenAI, Anthropic, or custom API key and model
- Coins -- Select which cryptocurrencies to trade (BTC, ETH, SOL, LTC, DOGE, LINK, XRP)
- Strategy -- contrarian (recommended), momentum, swing, or default

Your configuration is saved to arena\_config.json. Re-launch without the wizard:

```
uv run python launcher.py --config arena_config.json
```

### STEP 3 Live Trading (Real Money)

To trade with real money, select "Live" mode in Step 1 of the wizard and enter your Coinbase CDP credentials.

- Go to <https://portal.cdp.coinbase.com/access/api> and create an API key with trade permissions
- You will receive an API Key Name (organizations/...) and an API Secret (EC private key)
- Enter these when prompted by the launcher wizard

**WARNING: Live mode places REAL orders on Coinbase with real money. Start with a small balance.**

### STEP 4 Stop the Arena

Press Ctrl+C in the launcher window. To fully stop Kafka:

```
uv run python launcher.py --teardown
```

### Optional: Build Standalone .exe

Create a standalone executable that doesn't require Python or uv to be installed:

```
build_windows.bat
```

Output: dist\arena.exe -- Run it the same way as the Python launcher. Docker is still required for Kafka.

### Troubleshooting

| Problem             | Solution                                                          |
|---------------------|-------------------------------------------------------------------|
| "uv" not recognized | Close and reopen PowerShell after installing uv                   |
| Docker not running  | Open Docker Desktop and wait for it to fully start                |
| Port 9092 in use    | Run: <code>uv run python launcher.py --teardown</code>            |
| Connection refused  | Wait 30 seconds after Kafka starts -- it needs time to initialize |